Basic Mathematical Definition System of Self-Entanglement Duality Dynamics SED

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May 25, 2026

Abstract

This paper constructs a complete mathematical framework for Self-Entanglement Duality Dynamics (SED). Based on 2-category theory combined with self-referential recursion logic, it establishes nested duality structure, hierarchical self-entanglement, dual-driven dynamics, higher-order simplicial complex topology, background-free emergence equations, embedding relation between theoretical hierarchies and hierarchical transition operators. The whole system satisfies axiom self-consistency, and is compatible with mainstream physical duality theories, providing a basic mathematical foundation for the evolution of physical entities and dynamic generation of physical laws.

1 High-Order Category Framework: Construction of Nested Duality Space

1.1 Basic Definition of Second-Order Duality Category

SED² adopts a 2-category structure and is divided into two nested levels.

1.1.1 Object Hierarchy

- 0-order object $\mathcal{O}_0 = \{X, Y, Z, \dots\}$: physical entities, including spacetime points, matter fields and quantum states.
- 1-order object $\mathcal{O}_1 = \{\mathcal{R}_1, \mathcal{R}_2, \dots\}$: physical rules, including physical laws and dual evolution paradigms.

1.1.2 Morphism Hierarchy

- 1-morphism $f : X \to Y$: dual correlation between entities, representing the unity of opposites such as spacetime-matter and quantum-gravity.
- 2-morphism $\alpha : \mathcal{R}_i \Rightarrow \mathcal{R}_j$: dual transformation between rules, representing paradigm conversion between continuity-discreteness and classical-quantum.

1.1.3 Duality Lifting Functor

Any entity duality can generate corresponding rule duality and satisfy commutative diagram constraints:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\mathcal{R}_X"] & Y \arrow[d, "\mathcal{R}_Y"] \\ \mathcal{R}_X \arrow[r, "\mathcal{D}(f)"] & \mathcal{R}_Y \end{tikzcd}$$

$\mathcal{D}$ denotes the duality lifting functor, which realizes the mapping transformation from entity correlation to rule correlation.

1.2 Mathematical Definition of Hierarchical Self-Entanglement

1.2.1 First-Order Self-Entanglement

Non-equivalent mapping loop inside a single entity, distinguished from trivial identity transformation:

$$\mathcal{E}_1(X) := \{(f, g) \,\big|\, g \circ f \neq \mathrm{id}_X,\ f \circ g \neq \mathrm{id}_X\}$$

1.2.2 Second-Order Self-Entanglement

Self-interacting entanglement structure formed between rule paradigms:

$$\mathcal{E}_2(\mathcal{R}) := \{(\alpha, \beta) \,\big|\, \beta \circ \alpha \neq \mathrm{id}_{\mathcal{R}},\ \alpha \circ \beta \neq \mathrm{id}_{\mathcal{R}}\}$$

1.2.3 Hierarchical Binding of Entanglement

There exists a natural transformation $\eta : \mathcal{E}_1 \Rightarrow \mathcal{E}_2 \circ \mathcal{D}$, ensuring synchronous coordinated evolution of first-order and second-order entanglement.

1.3 Dual-Driven Dynamical Functor

1.3.1 Imbalance Measurement

First-order entity imbalance:

$$\Delta_1(f) = \|f \circ f^* - \mathrm{id}_Y\| + \|f^* \circ f - \mathrm{id}_X\|$$

Second-order rule imbalance:

$$\Delta_2(\alpha) = \|\alpha \circ \alpha^\dagger - \mathrm{id}_{\mathcal{R}_j}\| + \|\alpha^\dagger \circ \alpha - \mathrm{id}_{\mathcal{R}_i}\|$$

1.3.2 Two-Tier Driving System

The total driving functor is split into entity motion and rule evolution modules:

$$\mathcal{F} := \mathcal{F}_1 \oplus \mathcal{F}_2$$

$$\mathcal{F}_1(X) = X + \delta X, \quad \delta X \propto \nabla \Delta_1(f)$$

$$\mathcal{F}_2(\mathcal{R}) = \mathcal{R} + \delta \mathcal{R}, \quad \delta \mathcal{R} \propto \nabla \Delta_2(\alpha)$$

2 Entanglement Topology: Structure of Higher-Order Simplicial Complex

2.1 First-Order Entanglement Network

The first-order complex K_1 takes physical entities as basic units:

- 0-simplex: independent physical entities
- 1-simplex: first-order dual relation between entities
- 2-simplex: closed loop of many-body entanglement

Betti number $b_n(K_1)$ is used to describe topological characteristics, characterizing system connectivity and entanglement degrees of freedom.

2.2 Second-Order Entanglement Network

The second-order complex K_2 expands the rule dimension on the basis of the first-order structure. Newly added elements include rule objects, rule dual relations, rule entanglement loops, and 3-simplex of entity-rule coupling.

2.3 Hierarchical Conservation Law

The total Euler characteristic remains constant during system evolution:

$$\sum_n (-1)^n b_n(K_1) + \lambda \sum_m (-1)^m b_m(K_2) = \chi_{\text{total}} = \text{constant}$$

λ is the hierarchical coupling constant, describing the interaction strength between entity layer and rule layer.

2.4 Self-Interaction Property of Entanglement

The second-order self-interaction operator is defined to complete internal operation of rules, satisfying Jacobi identity. Topological dimension theorem: the dimension of the second-order complex is no less than the dimension of the first-order complex plus one.

3 Background-Free Emergence Equation

3.1 Hierarchical Emergence Rule

3.1.1 First-Order Emergence: Spacetime and Matter

Spacetime manifold and matter field are generated by statistical evolution of underlying entanglement network:

$$\mathcal{M} \times \Phi = \lim_{\epsilon \to 0} \frac{\mathrm{Tr}_{K_1}\left[e^{-\beta \hat{H}_1} \hat{\mathcal{O}}_{\mathcal{M}} \otimes \hat{\mathcal{O}}_{\Phi}\right]}{Z_1(\beta)}$$

3.1.2 Second-Order Emergence: Physical Laws

Physical laws naturally generate from higher-order structures based on self-consistent fixed point condition:

$$\mathcal{L} = \lim_{\Lambda \to \infty} \mathrm{Fix}\left(\mathcal{R} \mapsto \mathcal{F}_2(\mathcal{R}; K_2, \Lambda)\right)$$

3.2 Hierarchical Covariant Dynamics

3.2.1 Composite Covariant Derivative

Integrate derivatives of entity layer, rule layer and hierarchical coupling field:

$$\mathbb{D}_\mu := \nabla_\mu^{(1)} \otimes \mathbb{I}_2 + \mathbb{I}_1 \otimes \nabla_\mu^{(2)} + \mathcal{A}_\mu$$

3.2.2 General Field Evolution Equation

$$\mathbb{D}^\mu \mathbb{D}_\mu \Psi + \mathcal{V}'(\Psi) = \mathcal{J}_{\text{ent}}$$

Ψ denotes combined field quantity of entities and rules, $\mathcal{V}$ denotes hierarchical potential energy, and $\mathcal{J}_{\text{ent}}$ denotes entanglement source term.

3.2.3 Separate Evolution Equation of Rules

It consists of two parts: self-action of rules and bidirectional feedback of entity states:

$$\frac{\delta \mathcal{R}}{\delta t} = \int \mathcal{D}\alpha \, W[\alpha] \frac{\delta^2 \mathcal{R}}{\delta \alpha^2} + \int d^4x \sqrt{-g} \, T_{\mu\nu}[\Phi] \frac{\delta \mathcal{R}}{\delta g_{\mu\nu}}$$

3.3 Bidirectional Causal Closed Loop

Causal mapping satisfies self-consistent closed-loop condition, breaking the restriction of single one-way time sequence:

$$\mathcal{C} \circ \mathcal{C}^{-1} = \mathbb{I} + \mathcal{K}$$

System rules can complete self-verification and correction through causal retroactivity.

4 Formal Embedding Proof of SED^1 into SED^2

4.1 Rule-Frozen Subspace

Constraint conditions are set to lock the degrees of freedom of rule evolution:

$$\nabla_\mu^{(2)} \mathcal{R} = 0, \quad \hat{\Gamma}[\mathcal{R}] = 0, \quad \hat{\Lambda}[\Phi] = 0$$

Structures satisfying the conditions constitute the frozen submanifold $\mathcal{M}_{\text{freeze}}$.

4.2 Core Embedding Theorem

There exists a faithful full functor that can completely incorporate the first-order theory into the second-order theoretical framework:

$$\mathcal{I} : \mathbf{SED}^1 \hookrightarrow \mathcal{M}_{\text{freeze}} \subset \mathbf{SED}^2$$

4.3 Degradation Result under Limit

After rules are frozen, equations of the second-order system degenerate into basic equations of first-order Self-Entanglement Duality Dynamics:

$$\nabla^{(1)\mu} \nabla_\mu^{(1)} \Phi + V_1'(\Phi) = J_{\text{ent}}^{(1)}$$

The rule quantity keeps constant, and hierarchical conservation is simplified to single entity topological conservation.

4.4 Reserved Interface for Hierarchical Transition

Transition operator:

$$\hat{\mathcal{T}}_{1\to 2} = \exp\left(i \int d^4x \, \epsilon(x) \hat{\mathcal{O}}_{\text{break}}(x) \right)$$

Critical criterion: when the first-order entanglement complexity exceeds the critical threshold, frozen rules are automatically released, and the system completes phase transition from first-order to second-order.

5 Symbol Glossary

Symbol	Meaning
$\mathbf{SED^2}$	Category of Self-Entanglement Duality Dynamics
$\mathcal{O}_0, \mathcal{O}_1$	Sets of entity objects and rule objects
$\mathcal{D}$	Duality lifting functor
$\mathcal{E}_1, \mathcal{E}_2$	First-order and second-order self-entanglement structure
K_1, K_2	First-order and second-order entanglement simplicial complex
b_n, χ_{total}	Betti number and total Euler characteristic of system
$\mathbb{D}_\mu$	Composite hierarchical covariant derivative
$\mathcal{M}_{\text{freeze}}$	Rule-frozen submanifold
$\mathcal{I}$	Embedding functor from first-order to second-order system
$\hat{\mathcal{T}}_{1\to 2}$	Hierarchical phase transition operator

6 Theoretical Self-Consistency Verification

1. Nested duality origin: matched with two-layer object and morphism structure of 2-category
2. Second-order self-entanglement symbiosis: realized objectification of relations and high-order self-interaction
3. Dual mutual-driven evolution: split power sources via two-layer driving functor
4. Background-free hierarchical emergence: generate spacetime and laws based on fixed point equation
5. Bidirectional causal duality: construct closed-loop causal mapping system
6. Hierarchical covariant conservation: topological invariants satisfy conservation constraints in whole evolution

The theory is compatible with mainstream physical duality systems including T-duality, S-duality and Hodge duality, with no contradiction in basic logic.

7 Subsequent Derivation Directions

1. Operatorize entanglement structure and sort out algebraic operation relations
2. Establish sheaf theory modeling to unify global laws and local phenomena
3. Construct symplectic geometric framework and transform into Hamiltonian dynamical system
4. Strictly prove the existence of solutions for fixed point equation of rules
5. Derive renormalization group flow function and analyze critical characteristics of system phase transition